Quality gates before the solver: remaining duplicates, topic size, coverage of the syllabi, stability

Solid border: exists (prototype or in use)
Dashed border: proposed

Course corpus
syllabi and course facts

From courses to topics

1 · Evidence preparation
(no LLM)

2 · Per-course topic extraction
(LLM, each course on its own)

Raw topics
with evidence and attributes

3 · Canonicalisation
(embeddings + LLM)

4 · Topic review
(human)

Canonical topics and
raw → canonical mapping

5 · Re-annotation against
canonical topics (LLM)

Course * topic coverage
with weights

6 · Derived inputs
(no LLM)

7 · Dependency proposal
(LLM on plausible pairs)

Dependency review
(human)

Consensus communities over co-teaching
and description embeddings

Missing topic affinity means anti-affinity;
could be mandatory or weighted

Topic affinity analysis

Each topic: key, description, ECTS, level, kind,
technology, and the topics it depends on

Solver inputs

Prefixed modules

General module-generation
constraints

Topic-topic affinity

Topics

Solver

Solver outputs

Modules proposal

Outliers

Outliers detection

Human review